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Startups' demand for non-financial resources: Descriptive evidence from an international corporate venture capitalist

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ABSTRACT

Using proprietary data from Europe and Latin America, we show how startup characteristics shape their demand for non-financial resources from their corporate venture capitalists. We first highlight the investors' commercial network, investors' ties to other investors and their contacts to retail clients as the most important non-financial resources for early-stage startups. Second, we uncover systematic differences in startups' demand for non-financial resources depending on startups' slack financial resources, their business model, and size. Overall, we provide direct evidence on startups' demand for non-financial resources which is hard to detect and to quantify for researchers based on public data.

"The only mistake you can make is not asking for help."

Sandeep Jauhar, U.S. author

1. Introduction

Young startups are key drivers of innovation and economic growth, turning the question of performance-influencing factors into a major challenge for researchers, managers, and economists. The literature stresses the access to financial (Beck and Demirgüç-Kunt, 2006; Long, 2019) and non-financial resources (Alvarez-Garrido and Dushnitsky, 2016) as crucial parameters for startup success. However, we lack comprehensive evidence on the nature of the needed resources from the startups' perspective because startups usually seek and receive non-financial resources in private transactions. Therefore, researchers frequently have to make assumptions or rely on indirect evidence about the startups' demand for these resources (Kerr et al., 2014). Nevertheless, economists, policy makers, and investors need a better understanding of the startups' needs for non-financial resources to develop and promote a sound startup ecosystem.

In this study, we fill the literature gap by exploring the resource demand from early-stage high-tech startups in Europe and Latin America. We use proprietary data of 360 startups in 12 countries from a major corporate venture capitalist (CVC) which supports all startups with the same financial assets but a varying amount in non-financial resources. In the analysis, we condition the startups' demand for non-financial resources on different startup specific characteristics, namely the startups' amount of slack financial resources, their business model, and size.

Our findings underscore the importance of networks, connections to potential investors and retail clients as important non-

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financial resources which startups seek from investors in their early development stage. By documenting large differences in the startups' demand for non-financial resources, this study complements the literature on the consequences of external financing for startup growth (Beck and Demirgüç-Kunt, 2006; Long, 2019). We show that the startup-investor relationship extends far beyond the supply of financial resources, but also rests on non-financial CVC resources.

2. Financial slack and non-financial resources

Startups are typically resource-constrained (Hvide and Møen, 2010) and depend on external financial and non-financial resources. Frictions in the access to finance – so-called financial constraints – strongly restrict startup growth (Allen et al., 2013; Beck and Demirgüç-Kunt, 2006; Farinha and Félix, 2015; Long, 2019). Startups without sufficient financial resources cannot benefit from all profitable investment projects and, in turn, do not turn into high-growth firms (Bottazzi et al., 2014; Long, 2019).

Besides banks, strategic long term investors such as venture capitalists (VCs) are major entities in helping startups to overcome their resource constraints (see Da Rin et al., 2013; Manigart and Sapienza, 2017). Most VCs are established as independent entities with disperse ownership while few of them get sponsored by large and resource-rich corporations, the so-called CVCs, or governments (Riyanto and Schwienbacher, 2006; Tykvová, 2018). Startups benefit from VCs' financial but also non-financial resources, such as the assistance in building networks (e.g., Baum et al., 2000), the advice in internal processes, e.g. in human resources and legal issues (e.g., Dushnitsky and Lenox, 2005), or marketing (e.g., Anderson et al., 2018; Homburg et al., 2014). The provided non-financial resources help startups to reduce time as well as capital investments because, e.g., startups can exploit the CVCs' knowledge and infrastructure, and get faster access to the market (Alvarez-Garrido and Dushnitsky, 2016). Furthermore, they can focus on their core business and take advantage of the positive signaling effect from the cooperation with the CVC (Riyanto and Schwienbacher, 2006). Baum et al. (2000) summarize alliances "to be particularly beneficial to young, resource-constrained firms with established firms […] to overcome liabilities of newness and/or smallness".

Financial constraints also impact startups' demand for non-financial resources. The resource dependence perspective predicts that financial slack buffers startups' demand for resources that they can easily buy (Patzelt et al., 2008). They consequently demand "harder-to-replace" non-financial resources. On the contrary, financially constrained startups cannot convert financial into non-financial resources and, therefore, demand the most vital resources without replacement considerations.

3. Data and measures

We use unique, proprietary information from the Management Information System (MIS) by one CVC[1] within the information and communication technology sector. The CVC is a legally separate subsidiary of a globally acting European corporation which is listed in the EURO STOXX 50 Index. It operates offices in 12 countries in Europe and Latin America.[2] All startups are engaged with this CVC and receive the same contract with a one-time fixed financial payout but a varying amount and composition of non-financial resources.

Data is collected by the CVC within regularly conducted startup interviews. Therein, startups indicate the key business area in which they need the CVC's non-financial assistance. The seven possible areas are fundraising activities, commercial connections to the parent corporation, network to other associated startups, marketing, human resources, legal advice and product/operations. Additionally, we consider static startup characteristics as well as evolving information on startups' current business activities. Data is available for ten quarters in the years 2014 to 2016. The sample consists of 360 startups with 1215 observations. Sample descriptions regarding the local distribution of startups are provided in Table 1.

We measure the startups' financial constraints using the CVC's liquidity rating[3] which is periodically assigned to each startup. It is based on private information, especially startups' available cash and current cash burn rate. Ratings are ten-point scaled with higher ratings characterizing better equipped startups. Thereby, a rating of ten refers to startups with sufficient cash to cover 18 months at the current burn rate whereas a rating of zero refers to startups without enough cash for six months at the current burn rate. By using this internal measure for financial constraints, we avoid pitfalls from other proxies (Musso and Schiavo, 2008). We classify startups with a liquidity rating at and below the 25% quantile per country as financially constrained (30.70% of the observations).

Looking at the startups' business model, 52.35% follow a pure business-to-business (B2B) approach, 17.04% focus on retail customers (B2C) only, and 30.62% target both, businesses and retail customers (B2B & B2C). Startups show an average headcount of 7.95 and receive a mean liquidity rating of 5.12 (out of 10) from the CVC.

4. Results

In the following, we first descriptively display the startups' demand for different non-financial resources to document the startups' needs and characteristics which influence their resource demand. Subsequently, we conduct multivariate regressions including these

[1] Other aspects of the data are already used by Riepe et al. (2019).

[2] During the observation period (2014-2016), major economic turbulences in Venezuela occurred. We exclude Venezuelan observations from the sample to alleviate concerns that these dynamics are somehow unique which might drive our results.

[3] Other ratings cover the startups' current situation regarding 3rd party funding, commercial status, the startup's competitive situation, market opportunities, product quality, revenues and team quality.

Table 1
Local Distribution of Startups and Observations.

	Startups (N = 360)		Observations (N = 1215)	
	Freq.	%	Freq.	%
Latin America	**191**	**53.06**	**749**	**61.65**
- Argentina	42	11.67	165	13.58
- Brazil	17	4.72	39	3.21
- Chile	32	8.89	155	12.7
- Colombia	35	9.72	161	13.25
- Mexico	37	10.28	154	12.67
- Peru	28	7.78	75	6.17
Europe	**169**	**46.94**	**466**	**38.35**
- Czech Republic	15	4.17	52	4.28
- Germany	21	5.83	62	5.10
- Ireland	26	7.22	71	5.84
- Spain	54	15.00	143	11.77
- United Kingdom	53	14.72	138	11.36

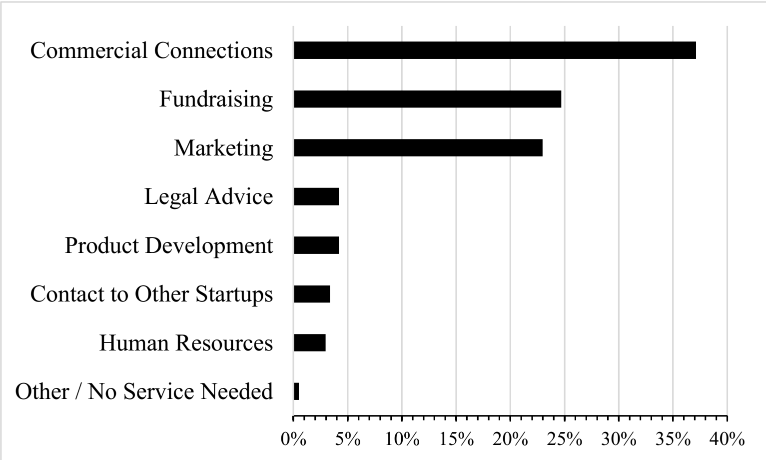

Fig. 1. Startups' Demand for Non-Financial Resources.

key characteristics to uncover systematic differences within the demand structure between groups. All values displayed in Fig. 1 to 4 in Section 4.1 are reported in Table 1A–3A of the Appendix A.

4.1. Descriptive evidence

4.1.1. Aggregated demand for non-financial resources

Fig. 1 shows the startups' responses to the question: "Which area can [the CVC] help you with?". With 37.12% of our observations, startups most often ask for support in establishing commercial connections to the CVC's parent corporation. As the true motives behind these requests, startups specify in the subsequent open question "to get closer to potential clients", "to achieve new partnerships around the world" and "to get commercial connections to pilot the product".[4] Results indicate startups' early focus on obtaining access to diverse information and a broad set of capabilities from the parent corporation at low costs. Our finding is in line with Baum et al. (2000) who highlight the value of alliances and information spillovers at the beginning of startups' lives.

Second, startups demand CVC's support in fundraising activities (24.69%). In the open question, startups underscore that they wish the CVC "to introduce some potential investor who fit with [the startup's] project", "to help by connecting [the startup] to credible VCs with specialized experience", or "to find and endorse the company to interested angel investors". One startup even explicitly states needing "help with contacts for our next funding round and orientation on how to not fuck it up". These answers highlight the central role of the CVC's network in generating and credibly conveying soft information to potential investors, making the CVC to perform similar functions as traditional financial institutions (Levine, 2005).

Fig. 1 also reveals the essential role of marketing activities for startups (22.96%). In the open question, startups clearly state their need for "better contacts to retail clients", "press connections both print and TV" and "marketing strategies or ideas to make people

[4] Similar answers were given over 100 times.

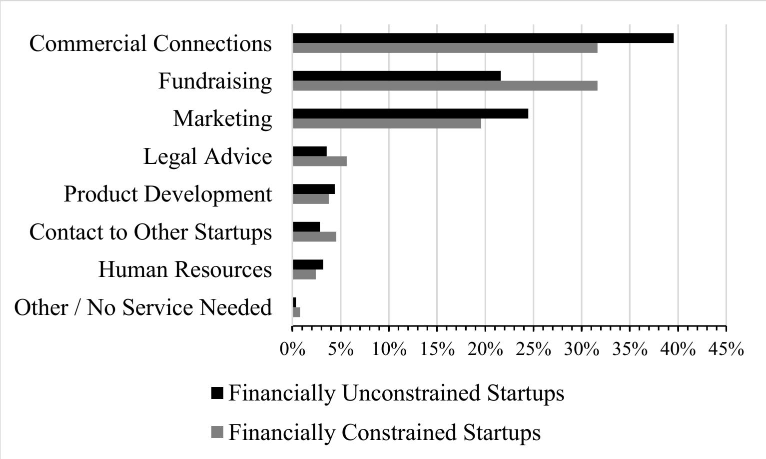

Fig. 2. Startups' Demand for Non-Financial Resources by Financial Constraints.

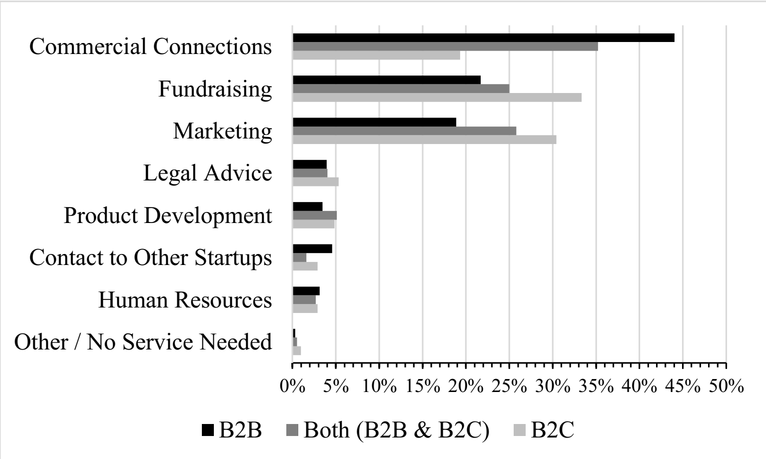

Fig. 3. Startups' Demand for Non-Financial Resources by Startups' Business Model.

get to know [the startup] and start [its product]". In line with Anderson et al. (2018) and Homburg et al. (2014), we find the support in advertising and marketing activities to appear highly relevant to early-stage startups. Results are also consistent with the role of the CVC and its parent corporation in linking their own and startups' market activities in a strategic way (Riyanto and Schwienbacher, 2006).

Other non-financial resources, such as legal advice, support with human resource processes or with regard to product development appear to be less relevant to high-tech startups. Fig. 1 indicates that less than 5% of the answers relate to the support in any of these areas.

4.1.2. Demand for non-financial resources by financial constraints, business model and size

Fig. 2 compares startups' demand for non-financial resources with higher and lower slack financial resources. Because startups can partially substitute financial and non-financial resources, we expect differences in startups' demand for non-financial resources. Fig. 2 reveals a lower demand for commercial connections and for marketing support by financially constrained startups. At the same time, financially constrained startups more heavily rely on the CVC's network with other investors to improve their fundraising activities.

Fig. 3 displays the startups' demand conditioned on the startups' business model. Startups engaged in B2B seek commercial connections (and contacts to other startups) significantly more often than those engaged in B2C. For B2C startups, the CVC's investor network and its marketing resources appear more important.

Fig. 4 displays the startups' demand for non-financial resources conditioned on the startups' size, measured by startups' full-time employees. It reveals the high demand for commercial connections from startups of all sizes. Nevertheless, there is a slightly stronger demand for commercial connections and marketing support from larger startups with more than nine employees whereas smaller startups more often demand help in their fundraising activities. Because especially smaller startups are more opaque and face higher information asymmetries vis-à-vis external investors, results are consistent with the CVC's function to overcome these obstacles and ensure startups' access to other investors.

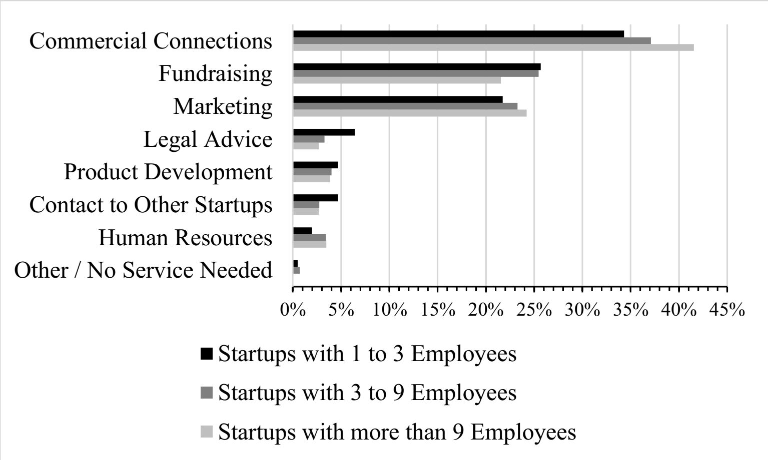

Fig. 4. Startups' Demand for Non-Financial Resources by Startup Size.

4.2. Multivariate regression

Table 2 underscores our descriptive analysis from Section 4.1. In the multivariate regression, we find statistically significant differences in the requests for commercial connections and fundraising between financially unconstrained and financially constrained startups. While the former group requests significantly more help in commercial connections, the latter significantly more often asks for non-financial help in fundraising. Table 2 also reveals the tendency of financially unconstrained startups to request assistance in marketing.

Our results are consistent with the essential function of conveying hard and soft information within the investors' network through the CVC. Furthermore, it is consistent with a higher focus of the managerial team on fundraising activities by financially constrained startups. Although financial constraints can foster using resources efficiently (Hvide and Møen, 2010), our results highlight potential downsides. As managerial attention and the demand for non-financial resources shift towards external fundraising, financial constraints might prevent startups from acquiring non-financial resources in client and growth-related areas.

Furthermore, Table 2 shows that B2C startups significantly more often seek help in fundraising and marketing, whereas B2B startups request assistance in commercial connections. This enables the latter group to easier form alliances (Baum et al., 2000; Patzelt et al., 2008). Results are consistent with a stronger focus by B2B startups on their networks, whereas B2C startups use the CVC's non-financial resources to foster their relationship to retail clients. In line with the descriptive evidence from Fig. 4, we do not find significant differences in the requests for non-financial resources when splitting by startup size. However, regression results again indicate the slight tendency of larger startups to ask for help in commercial connections compared to smaller ones.

Table 3 displays the results from the likelihood ratio tests using the main three non-financial resources. It emphasizes the significance of financial constraints and the startups' business model regarding their request for non-financial resources. On the contrary, startup size does not seem to be important regarding the startups' requests.

Table 2
Effect on the Main Non-Financial Resources.

DEPENDENT VARIABLES:	(1) Commercial Connections	(2) Fundraising	(3) Marketing
Financial Constraints	**−0.247***** **(0.0042)**	**0.333***** **(0.0002)**	−0.122 (0.193)
B2C	**−0.780***** **(0.000)**	**0.370***** **(0.0009)**	**0.405***** **(0.0004)**
Both (B2B & B2C)	**−0.251***** **(0.0037)**	0.137 (0.147)	**0.267***** **(0.0049)**
Medium Size	−0.0054 (0.952)	0.0651 (0.494)	0.0468 (0.630)
Large Size	0.156 (0.178)	−0.0240 (0.848)	0.00490 (0.969)
Observations	1215	1215	1215
Chi^2	92.91	64.06	62.54
Pseudo R^2	0.0580	0.0472	0.0478
Country and Time FE	Yes	Yes	Yes
Constant	Yes	Yes	Yes

P-values are displayed in parentheses. ***, **, and * represent significance at the 1%, 5%, and 10% levels.

Table 3
Likelihood Ratio Tests.

	Chi^2	Degrees of Freedom	p-value
Commercial Connections			
Financial Constraints	8.2449	1	0.0041***
Business Model	48.3368	2	< 0.0001***
Size	2.6221	2	0.2695
Fundraising			
Financial Constraints	14.1653	1	0.0002***
Business Model	11.0066	2	0.0041***
Size	0.8598	2	0.6506
Marketing			
Financial Constraints	1.7039	1	0.1918
Business Model	15.5416	2	0.0004***
Size	0.2921	2	0.8641

***, **, and * represent significance at the 1%, 5%, and 10% levels.

5. Discussion and conclusion

Startups are of crucial importance for innovation and economic growth. In order to properly fulfill this role, they rely on external help in financial as well as non-financial resources, especially in their early development stage. This study adds to the literature on early-stage startups by using proprietary data of startups' hard-to-measure resource demand. We provide descriptive evidence on the non-financial resources which startups seek from their CVC and find significant differences depending on startup specific characteristics, such as the startups' financial situation, their business model, and size. This underscores the needs of early-stage startups in non-financial aspects. We complement prior studies by highlighting descriptively the importance of non-financial resources for startup performance.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2019.101321.

Appendix A

Table 1A–Table 3A.

Table 1A
Startups' Demand for Non-Financial Resources and split by Financial Constraints.

	All Startups	By Financial Constraints (1) Unconstrained (N = 842)	(2) Constrained (N = 373)
Startups demand help in			
Commercial Connections	37.12%	**39.55%**	**31.64%**
Fundraising	24.69%	**21.62%**	**31.64%**
Marketing	22.96%	**24.47%**	**19.57%**
Legal Advice	4.20%	**3.56%**	**5.63%**
Product Development	4.20%	4.39%	3.75%
Contact to Other Startups	3.37%	2.85%	4.56%
Human Resources	2.96%	3.21%	2.41%
Others / No Service Needed	0.49%	0.36%	0.80%

Bold figures indicate statistically significant differences at the 10% level.

Table 2A
Startups' Demand for Non-Financial Resources split by Business Model.

	By Business Model (1) B2B (N = 636)	(2) Both (B2B & B2C) (N = 372)	(3) B2C (N = 207)	Differences (1) → (2)	(2) → (3)	(1) → (3)
Startups demand help in						
Commercial Connections	44.03%	35.22%	19.32%	**−8.81%**	**−15.89%**	**−24.70%**
Fundraising	21.70%	25.00%	33.33%	3.30%	**8.33%**	**11.64%**
Marketing	18.87%	25.81%	30.43%	**6.94%**	4.63%	**11.57%**
Legal Advice	3.93%	4.03%	5.31%	0.10%	1.28%	1.38%
Product Development	3.46%	5.11%	4.83%	1.65%	−0.28%	1.37%
Contact to Other Startups	4.56%	1.61%	2.90%	**−2.95%**	1.29%	−1.66%
Human Resources	3.14%	2.69%	2.90%	−0.46%	0.21%	−0.25%
Others / No Service Needed	0.31%	0.54%	0.97%	0.22%	0.43%	0.65%

Bold figures indicate statistically significant differences at the 10% level.

Table 3A
Startups' Demand for Non-Financial Resources split by Startup Size.

	By Size (Employees) (1) 1 to 3 (N = 405)	(2) > 3 to 9(N = 550)	(3) > 9 (N = 260)	Differences (1) → (2)	(2) → (3)	(1) → (3)
Startups demand help in						
Commercial Connections	34.32%	37.09%	41.54%	2.94%	4.45%	**7.39%**
Fundraising	25.68%	25.45%	21.54%	−0.10%	−3.92%	−4.01%
Marketing	21.73%	23.27%	24.23%	1.65%	0.96%	2.61%
Legal Advice	6.42%	3.27%	2.69%	**−3.12%**	−0.58%	**−3.70%**
Product Development	4.69%	4.00%	3.85%	−0.67%	−0.15%	−0.82%
Contact to Other Startups	4.69%	2.73%	2.69%	−1.94%	−0.03%	−1.98%
Human Resources	1.98%	3.45%	3.46%	1.49%	0.01%	1.50%
Others / No Service Needed	0.49%	0.73%	0.00%	−0.26%	−0.73%	−0.98%

Bold figures indicate statistically significant differences at the 10% level.

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